Six models. One benchmark.

6/6 completed runs verified locally. One frozen benchmark; validation-selected checkpoints. Observed test quality, with failures and model size kept visible.

HIGHEST TEST PSNR

32.58 dB

Restormer

HIGHEST TEST SSIM

0.9297

Restormer

FEWEST PARAMETERS

0.935 M

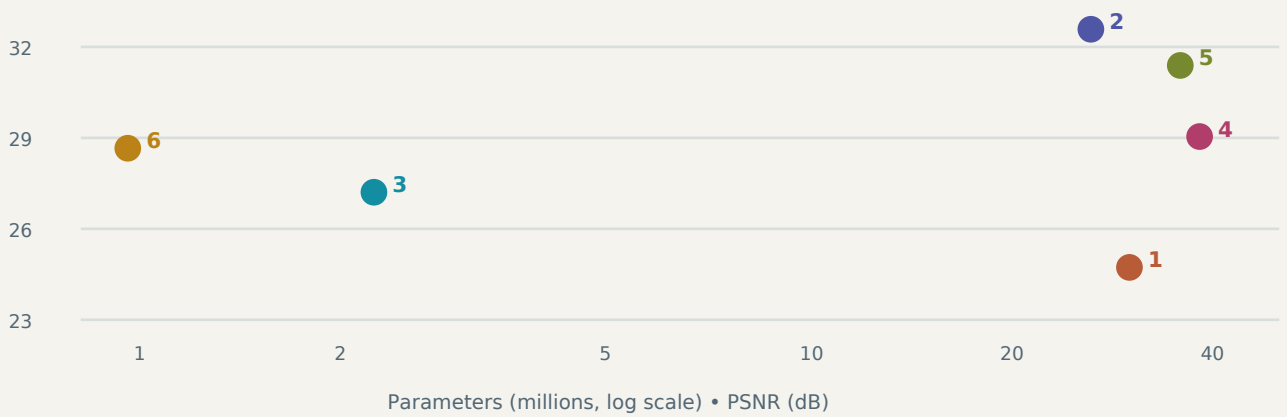
RAMiT

01 / OVERALL TEST QUALITY

Model	PSNR ↑	SSIM ↑	ΔPSNR ↑	ΔSSIM ↑	Params M ↓
Restormer	**32.58**	**0.9297**	**13.97**	**0.2162**	26.127
PromptIR	31.39	0.9182	12.77	0.2047	35.592
TransWeather	29.04	0.9000	10.43	0.1865	38.052
RAMiT	28.66	0.8995	10.04	0.1860	**0.935**
LiteWeather v2	27.21	0.8644	8.59	0.1509	2.189
Attention Pix2Pix	24.73	0.8201	6.11	0.1066	29.855

Read the table: green bold cells mark the best displayed value in each column; ties share highlighting. Deltas compare restored images with the same degraded inputs. Each weather contributes 25% of the headline score, regardless of image count.

02 / QUALITY VERSUS PARAMETER COUNT



1 Attention Pix2Pix · 2 Restormer · 3 LiteWeather v2 · 4 TransWeather · 5 PromptIR · 6 RAMiT

Where each model succeeds.

Per-weather scene-balanced metrics. A strong average can still hide damaged images.

03 / PSNR BY WEATHER — HIGHER IS BETTER

Model	Rain	Fog	Snow	Smoke
Attention Pix2Pix	28.46	23.55	29.05	17.85
Restormer	**38.77**	31.51	**39.51**	**20.55**
LiteWeather v2	31.02	27.42	31.40	18.98
TransWeather	34.19	29.58	32.31	20.10
PromptIR	36.40	**31.88**	37.63	19.65
RAMiT	34.37	24.19	36.85	19.23

04 / SSIM BY WEATHER — HIGHER IS BETTER

Model	Rain	Fog	Snow	Smoke
Attention Pix2Pix	0.8674	0.8708	0.8514	0.6909
Restormer	**0.9768**	0.9666	**0.9816**	**0.7937**
LiteWeather v2	0.8772	0.9353	0.9045	0.7408
TransWeather	0.9507	0.9525	0.9260	0.7710
PromptIR	0.9656	**0.9672**	0.9739	0.7662
RAMiT	0.9526	0.9206	0.9685	0.7561

05 / IMAGES MADE WORSE — PSNR / SSIM COUNTS ↓

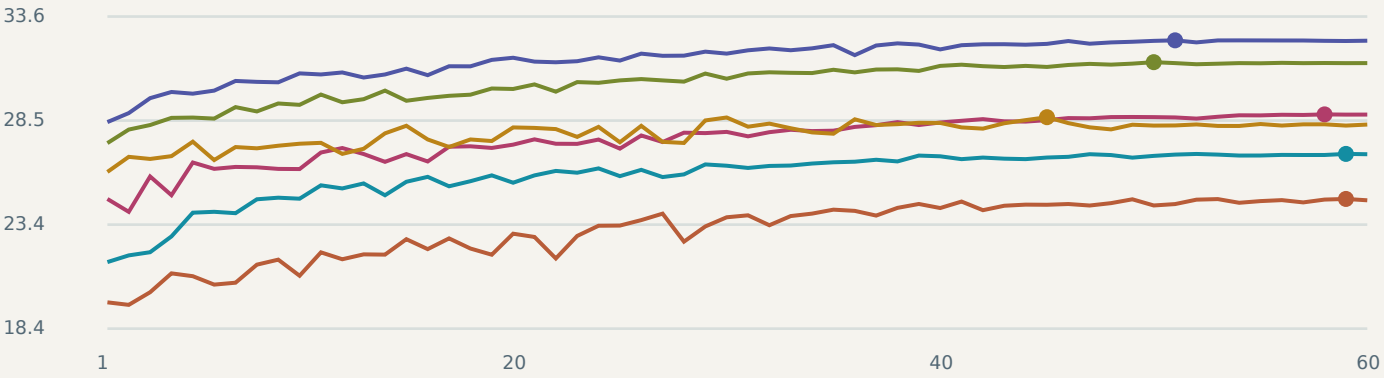
Model	Rain /224	Fog /1000	Snow /768	Smoke /342
Attention Pix2Pix	21 / 29	9 / 105	0 / 0	179 / 341
Restormer	0 / 1	0 / 0	0 / 0	27 / 37
LiteWeather v2	3 / 11	1 / 0	0 / 0	88 / 249
TransWeather	2 / 4	0 / 1	0 / 0	58 / 106
PromptIR	0 / 1	1 / 0	0 / 0	68 / 128
RAMiT	6 / 11	41 / 56	0 / 0	79 / 200

Smoke deserves separate scrutiny: 342 test images cover just six independent backgrounds. PSNR and SSIM regression counts overlap; do not add them together.

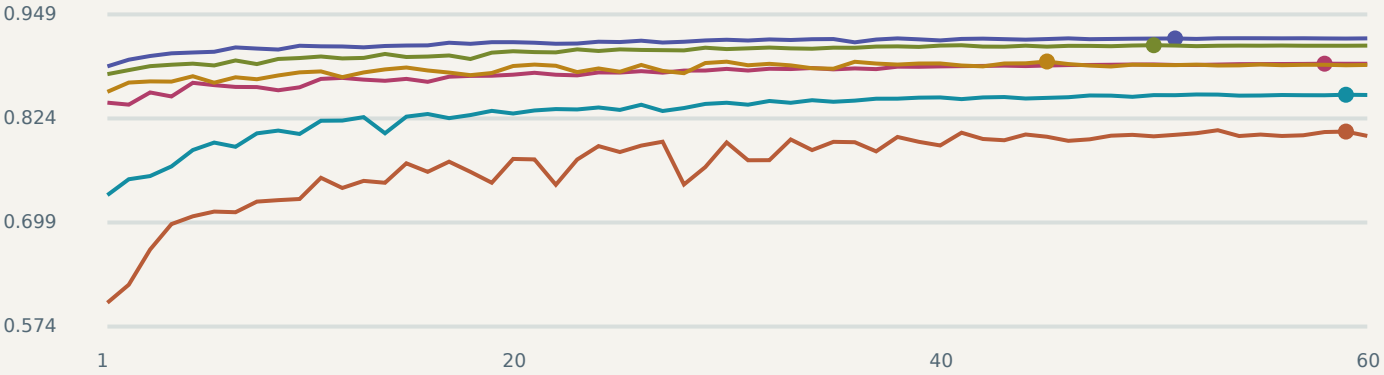
Learning curves. Real costs.

Every curve covers 60 epochs. Dots mark the checkpoint selected by validation PSNR, not by test scores.

Validation scene-balanced PSNR (dB)



Validation scene-balanced SSIM



Attention Pix2Pix · Restormer · LiteWeather v2 · TransWeather · PromptIR · RAMiT

06 / RECORDED TRAINING AND CHECKPOINT FOOTPRINT

Model	Best epoch	Epoch hours	best.pt MiB	Inference MiB
Attention Pix2Pix	59	1.45	373.6	114.0
Restormer	51	15.38	299.7	99.9
LiteWeather v2	59	1.64	25.3	8.4
TransWeather	58	2.73	433.7	145.3
PromptIR	50	16.67	406.4	136.0
RAMiT	45	13.92	12.0	4.4

Epoch hours sum logged training + validation time; exclude queues, preflight and final test. Parameter count and checkpoint size do not measure Jetson latency. INT8 quality, memory and FPS alongside YOLOv8 remain unmeasured.

The evidence behind the scores.

A reproducible protocol, an honest interpretation, and locally traceable weights.

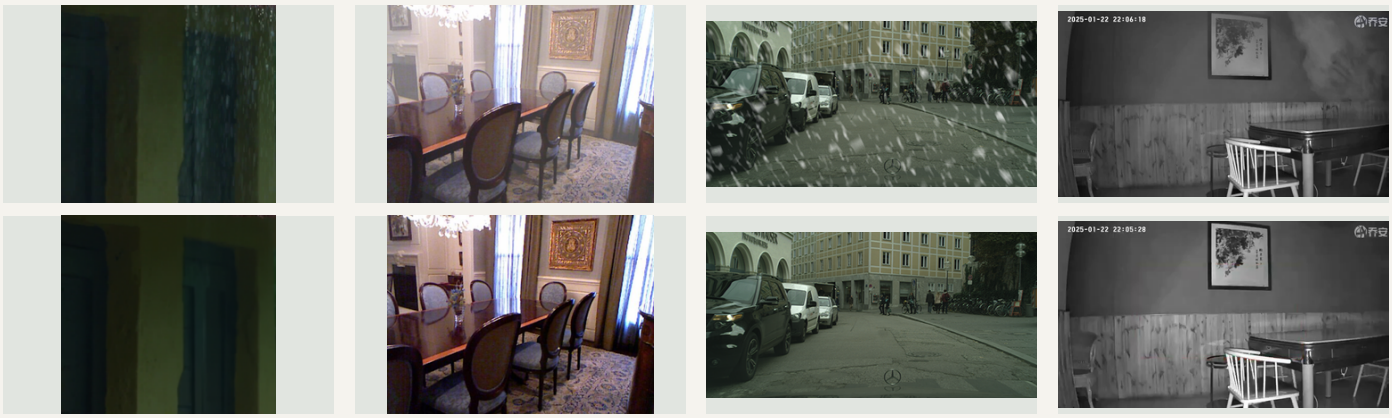
07 / FROZEN SPLIT

Weather / source	Train	Validation	Test
Rain / RealRain1k-H	773	112	224
Fog / RESIDE6K	5102	897	1000
Snow / SnowCityscapes	5382	618	768
Smoke / SmokeBench v2	7353	384	342
Total	18610	2011	2334

Shared sampling: seed 42; 4,096 draws/epoch with replacement; rain/fog/snow/smoke probabilities 0.30/0.30/0.30/0.10; batch 4; paired 256-pixel crops and flips. 60 epochs = 61,440 generator updates. Native-size BF16 validation and test. Headline scores average within strata, then backgrounds, then weather classes.

08 / EXAMPLE DEGRADED INPUTS AND CLEAN TARGETS

Rain | **Fog** | **Snow** | **Smoke**



Top: degraded input. Bottom: clean target. Fixed score-blind examples from the prior audit; lowest sample-ID SHA256 per weather among images ≤500,000 pixels. Display thumbnails only; metrics use native resolution. These are dataset pairs, not model predictions.

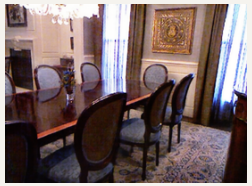
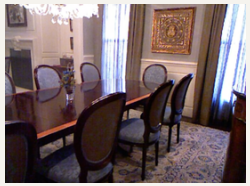
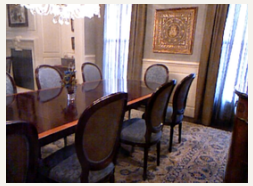
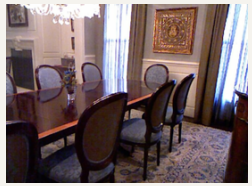
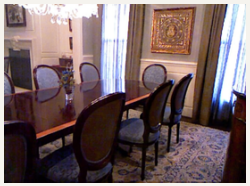
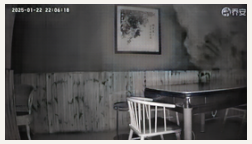
09 / WHAT THE COMPARISON CAN SUPPORT

Attention Pix2Pix and LiteWeather v2 started randomly. Restormer and RAMiT used deraining weights; TransWeather used All-Weather weights; PromptIR used all-in-one weights. Unequal pretraining means this is a system comparison, not an isolated architecture test. Pretraining overlap is not ruled out, particularly for RESIDE fog. Single-seed ranks and small score gaps do not establish statistical superiority. Cluster-bootstrap intervals are retained in the metrics JSON; they do not measure training-seed uncertainty.

Local handoff: artifacts/models/<model>/ contains best.pt, inference.pt, metrics, history and SHA256SUMS. Original run folders and earlier Restormer results remain preserved.

Same scene. Six restorations.

Fixed, score-blind examples • Native-resolution CPU FP32 inference • No cherry-picking by model scores

Degraded	Attention Pix2Pix	Restormer	LiteWeather v2	TransWeather	PromptIR	RAMiT	Clean target
RAIN							
							
FOG							
							
SNOW							
							
SMOKE							
							

These four examples illustrate behavior; they are not a representative visual survey. Numerical rankings use all 2,334 test images in BF16.

Full-resolution PNGs, sample IDs and checkpoint hashes are saved beside this PDF in examples/.